

COSC 301 Report

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Questions

Which Toronto neighbourhoods pose the greatest safety risk and does crime severity correlate with traffic collision severity at the neighbourhood level? Where and when crimes are likely to happen using predictive models?

1 Method

We used two major datasets from Toronto's Police Service Open Data:

- **Major Crime Indicators** (<https://open.toronto.ca/dataset/major-crime-indicators/>): Contains reported crimes across Toronto categorised into 6 major crime types. We examined `EVENT_UNIQUE_ID`, `OCC_DATE`, `NEIGHBOURHOOD_158`, `CSI_CATEGORY`, `OFFENCE`, `DEATH`, `INJURIES`, `EVENT_TYPE`, and `PREMISE_TYPE`.
- **Motor Vehicle Collisions involving Killed or Seriously Injured Persons** (<https://open.toronto.ca/dataset/motor-vehicle-collisions-involving-killed-or-seriously-injured-persons/>): Contains traffic collision records with features including `EVENT_UNIQUE_ID`, `OCC_DATE`, `FATALITIES`, `INJURY_COLLISIONS`, `AUTOMOBILE`, `MOTORCYCLE`, `PASSENGER`, `BICYCLE`, `PEDESTRIAN`, `NEIGHBOURHOOD_158`, `FTR_COLLISIONS`, and `PD_COLLISIONS`.

We used these datasets to rank neighbourhood safety and build predictive models for where and when incidents were likely to occur. We downloaded the datasets as CSVs and loaded them into a PostgreSQL database using Python. The crime data was cleaned with a multi-step pipeline before loading. The traffic data was loaded directly after adding time-based feature columns. Both tables were loaded row by row with date parsing and type conversion handled in Python before insertion. Both cleaned datasets were stored in PostgreSQL database with typed schemas. Safety scores were computed via SQL aggregation queries and exported as CSVs so they can be used in Power BI. We cleaned these datasets to answer our questions. For our tech stack, we used Python for scripting and feature engineering, Power BI for data visualization, and PostgreSQL for data storage.

1.1 Safety Score Calculation

For both datasets, we created safety scores which would help us rank neighbourhood safety.

1.1.1 Crime Safety Score

For the crime data (implemented in `open_consolidated_safety_score.py`), we calculated the safety score using:

$$\text{Safety Score} = \text{CSI_WEIGHT} \times \text{OFFENCE_WEIGHT} \times \text{normalized(Frequency)}$$

where

$$\text{Frequency} = \sum (\text{RECENCY_WEIGHT} \times \text{Frequency per year})$$

We assigned CSI Weights to each row based on the CSI category, with assault being the highest and non-MCI the lowest. Since some offences associated with CSI Weights are more severe than others, we also included an Offence Weight to ensure rows with more severe offences within a CSI category had more weight. For each row, we calculated crime Frequency/year by summing each year's RECENCY_WEIGHT multiplied by the frequency of the crime occurring in that year, then normalized the result on a scale from 0 to 1 to use as a weight. We multiplied all components to obtain each row's safety score. Finally, we ranked neighbourhoods in descending order from safest to most dangerous based on their safety scores.

1.1.2 Traffic Safety Score

For the traffic data (implemented in `traffic_safety_score.py`), we used:

$$\text{Raw_danger} = 5 \times \text{fatal_rate} + 3 \times \text{injury_rate} + 2 \times \text{ftr_rate} + 2 \times \text{vulnerable_rate} + 1 \times \text{pd_rate}$$

$$\text{Rate} = \frac{\text{recency-weighted count}}{\text{total recency-weighted collisions}} \times 1000$$

$$\text{Blended_danger} = 0.7 \times \text{danger_norm} + 0.3 \times \text{vol_norm}$$

$$\text{Safety Index} = 1 - \text{Blended_danger}$$

We first applied a recency weight to each year's collision counts using the same recency weights as the crime data's score, meaning collisions that happened in recent years have greater influence on the final score. Then, we calculated rates per 1000 recency-weighted collisions for five collision types: fatalities, injury collisions, fail-to-remain collisions, vulnerable road users, and property damage collisions. We assigned severity weights to each rate (fatalities highest at 5, property damage lowest at 1) and summed these to produce a raw danger score per neighbourhood. Since a neighbourhood with a high total collision count should be penalised even if per-collision rates are moderate, we blended 70% of the normalised danger score with 30% of the normalised collision volume. Finally, we inverted this blended danger score and scaled it to a 0-100 safety index (higher score indicates safer neighbourhood) and ranked all 158 neighbourhoods accordingly.

1.2 Predictive Models

Besides safety scores, we created four predictive models for traffic collisions, assault, auto theft, and break and enter using LightGBM, a decision tree algorithm. These models predict whether a crime type would occur in a neighbourhood within a fixed time window (3 days for collisions and assault, 7 days for auto theft and break and enter) for all neighbourhoods. Models were trained on 2014-2024 data and validated on 2025 data.

2 Results

For both crime and traffic data, we created maps indicating the safety score of each neighbourhood, coloured from green (safest) to red (most dangerous).

By examining crime results between 2020-2025, assault offences are the most common across many years. Scatterplots show a negative correlation between the number of selected incidents and safety score, which is expected: fewer crimes yield a higher safety score, and more crimes yield a lower safety score. Assault is the highest type of offence for most neighbourhoods regardless of safety score. Auto theft and break and enter are the two most common offences for neighbourhoods with stronger safety scores. In 2020, Moss Park had the most reported cases; from 2021-2025, West Humber-Claireville had the most reported cases.

For traffic results between 2020-2025, the trendline for collision types shows a decrease in 2020 due to the COVID-19 pandemic, but a notable increase through 2022, peaking in 2024. According to the map illustration and bar chart, Wexford/Maryvale is the neighbourhood with the lowest safety score and highest collision value, making it the most dangerous neighbourhood based on traffic collisions.

Based on crime safety scores, Moss Park and West Humber-Claireville are the two most dangerous neighbourhoods for crime. For traffic safety scores, Wexford/Maryvale is the most dangerous neighbourhood for traffic. Interestingly, West Humber-Claireville appears in the top rankings for both crime and traffic, suggesting it may have the highest overall safety risk. However, our safety scores for crime and traffic do not correlate, indicating the two scores are independent when analysing neighbourhoods.

3 Limitations

We chose the values of weights used for calculating safety scores rather than validating them against an external standard. Changes in weights would alter results and produce different neighbourhood rankings. There was a limited amount of data to train the models, which means the models have less data to learn from, reflected in their lower accuracy scores for less frequent crime types. The crime and traffic safety scores were calculated separately, meaning we did not measure the correlation between the two scores. Our data only includes crimes reported to the police department, so safety scores may not fully reflect the actual level of risk in every neighbourhood.

4 Next Steps

We can look for basic connections between crime types (e.g., theft vs. assault) and neighbourhood income levels using public census data to better understand socioeconomic issues associated with specific crimes. We can also integrate extra contextual data to improve model accuracy. Finally, we can explore other prediction models such as LSTMs, as we currently use LightGBM.

5 References

- Xu Zhang, Lin Liu, Minxuan Lan, Guangwen Song, Luzi Xiao, Jianguo Chen, Interpretable machine learning models for crime prediction, *Computers, Environ-*

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- Nicho, M., Hamed, A., Gaber, T., & Al Arimi, J. H. (2025). A-XGBoost: a resilient machine learning technique for predicting crimes against women across cultures on low cardinality crime data. *Cogent Social Sciences*, 11(1). <https://doi.org/10.1080/23311886.2025.2527392>
- GeeksForGeeks, (2021). <https://www.geeksforgeeks.org/machine-learning/lightgbm-vs-xgboost-which-algorithm-is-better/>
- Toronto Open Data, <https://open.toronto.ca/dataset/major-crime-indicators/>
- Toronto Open Data, <https://open.toronto.ca/dataset/motor-vehicle-collisions-involv>

A Annex: Images

The following figures are included to support the results presented in this report.

Figure 1: Crime Safety Score Map of Toronto Neighbourhoods (2020-2025)

Figure 2: Traffic Safety Score Map of Toronto Neighbourhoods (2020-2025)

Figure 3: Assault Offence Trends Over Time

Figure 4: Scatterplot: Safety Score vs. Number of Incidents

Figure 5: Traffic Collision Trends (2020-2025)

Figure 6: Top Dangerous Neighbourhoods for Crime and Traffic